Fourier Lemmas on the Three-Torus: Heat Smoothing, the L^2 Strain Identity, and the Curl of the Energy Derivative, Formalised in Lean 4

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DRAFT — 30 October 2026

Abstract

On $\mathbb{T}^3 = (\mathbb{R}/\mathbb{Z})^3$, the product of three circles, we prove in Lean 4 with Mathlib, with no `sorry` and only the standard axioms: the heat semigroup is the Fourier multiplier $e^{-\nu|n|^2 t}$, obeys the semigroup law and contracts L^2; it maps H^s to H^{s+k} with $\|e^{t\nu\Delta} f\|_{H^{s+k}}^2 \le k!\, e^{2\nu t}(2\nu t)^{-k}\|f\|_{H^s}^2$; for divergence-free velocity fields $\|S\|_2^2 = \frac{1}{2}\|\omega\|_2^2$; and the kinetic energy, viewed as a function of vorticity, has derivative the stream function, whose curl is the velocity.

1 Setting

$\mathbb{T}^3$ carries normalised Haar measure; $e_n(x) = e^{2\pi i n\cdot x}$, $n \in \mathbb{Z}^3$, is a Hilbert basis of $L^2(\mathbb{T}^3)$ (Mathlib's `mFourierBasis`), and $\hat{f}(n) = \int f\,\overline{e_n}$.

Lemma 1 (Parseval, `parseval_norm_sq`)**.** $\|f\|_2^2 = \sum_n |\hat{f}(n)|^2$.

Proof. The coefficient map is a linear isometry $L^2 \to \ell^2(\mathbb{Z}^3)$ and $\|a\|_{\ell^2}^2 = \sum |a_n|^2$. □

Lemma 2 (`norm_multiplier_le`)**.** *If $|m(n)| \le 1$ for all n, then $T_m f = \sum m(n)\hat{f}(n)e_n$ defines $T_m : L^2 \to L^2$ with $\|T_m f\|_2 \le \|f\|_2$.*

Proof. $\sum |m(n)\hat{f}(n)|^2 \le \sum |\hat{f}(n)|^2 < \infty$, so the sequence lies in ℓ^2; pull back by the isometry and apply Parseval. □

Commentary. The earlier manuscript already ran the analysis through Fourier coefficients: "The Fourier multiplier proof of the Sobolev embedding $H^s(\mathbb{T}^3, \mathbb{R}^3) \hookrightarrow C^1(\mathbb{T}^3, \mathbb{R}^3)$ for $s > 5/2$ follows from Parseval's identity and the p-series convergence" (*vFinal*, Drive). When this build was planned, the author's point was simply: "the torus is proved to be a Circle (cubic)" (30 October 2026). $\mathbb{T}^3$ is three circles, so Mathlib's multi-dimensional Fourier series and Parseval apply directly, and the Sobolev scale, missing from Mathlib, is defined here as a weighted coefficient sum.

2 Heat semigroup

Theorem 3. *For $\nu, s, t \geq 0$: $\widehat{e^{t\nu\Delta} f}(n) = e^{-\nu|n|^2 t}\hat{f}(n)$, $e^0 = \mathrm{id}$, $e^{s\nu\Delta}e^{t\nu\Delta} = e^{(s+t)\nu\Delta}$, $\|e^{t\nu\Delta} f\|_2 \leq \|f\|_2$.*

Proof. The symbol has modulus ≤ 1; $e^{-\nu|n|^2 s}e^{-\nu|n|^2 t} = e^{-\nu|n|^2(s+t)}$ and operators with equal coefficients are equal. □

Commentary. The local-existence block of the May drafts used exactly this operator: "The heat semigroup $e^{\nu t\Delta}$ is bounded on H^s ($s > 5/2$) with $\|e^{\nu t\Delta}\|_{\mathcal{L}(H^s)} \leq 1$" (*May 16 edits*, Drive). There it was asserted; here it is the contraction theorem above, for every s, since the symbol has modulus at most one at every mode.

3 Sobolev smoothing

$\|f\|_{H^s}^2 = \sum_n (1 + |n|^2)^s |\hat{f}(n)|^2$.

Lemma 4. *For $k \in \mathbb{N}$, $a > 0$, $x \geq 0$: $(1 + x)^k e^{-ax} \leq k!\, e^a a^{-k}$.*

Proof. $y^k/k! \leq e^y$ with $y = a(1 + x)$, and $e^{-ax} = e^a e^{-a(1+x)}$. □

Theorem 5 (`heat_smoothing`)**.** *For $\nu, t > 0$, $s \in \mathbb{R}$, $k \in \mathbb{N}$, $f \in H^s$: $e^{t\nu\Delta} f \in H^{s+k}$ and $\|e^{t\nu\Delta} f\|_{H^{s+k}}^2 \leq k!\, e^{2\nu t}(2\nu t)^{-k}\|f\|_{H^s}^2$.*

Proof. Mode by mode, $(1+|n|^2)^{s+k}e^{-2\nu t|n|^2}|\hat{f}|^2 \leq k!e^{2\nu t}(2\nu t)^{-k}(1+|n|^2)^s|\hat{f}|^2$ by the lemma with $a = 2\nu t$; sum by comparison. □

Theorem 6 (`heat_smoothing_real`)**.** *The same holds for every real $k \geq 0$ with $k!$ replaced by $\lceil k \rceil!$.*

Proof. For $y \geq 0$, $y^k \leq \lceil k \rceil!\, e^y$: if $y \leq 1$ then $y^k \leq 1 \leq e^y$; if $y \geq 1$ then $y^k \leq y^{\lceil k \rceil} \leq \lceil k \rceil!\, e^y$. The rest is as before. □

Commentary. The v2.4 master draft names this step as indispensable: "Viscosity is essential for local existence in H^s, the explicit higher-norm Gronwall estimate, and the final parabolic smoothing" (*NS Global Regularity v2.4 Master*, Drive, 25 May 2026). The drafts invoked the smoothing without a statement; the theorem above gives it with an explicit constant.

4 Strain and vorticity

For $k \in \mathbb{R}^3$, $\hat{u} \in \mathbb{C}^3$, $k \cdot \hat{u} = 0$: $\hat{S}_{ij} = \pi i(k_i \hat{u}_j + k_j \hat{u}_i)$, $\hat{\omega} = 2\pi i\, k \times \hat{u}$.

Lemma 7. *$\sum_{ij} |\hat{S}_{ij}|^2 = \frac{1}{2}|\hat{\omega}|^2$ and $|\hat{\omega}|^2 = 4\pi^2 |k|^2 |\hat{u}|^2$.*

Proof. Splitting $\hat{u} = a + ib$, for each real part $\sum_{ij}(k_i a_j + k_j a_i)^2 = 2(|k|^2|a|^2 + (k \cdot a)^2)$ and $|k \times a|^2 = |k|^2|a|^2 - (k \cdot a)^2$; both $k \cdot a$ and $k \cdot b$ vanish. □

Theorem 8 (`L2_strain_eq`)**.** *For L^2 fields with these coefficients, $\|S\|_2^2 = \frac{1}{2}\|\omega\|_2^2 \le \|\omega\|_2^2$.*

Proof. Parseval on each component, exchange the finite and infinite sums, apply the lemma per mode. □

5 The curl of the energy derivative

$\hat{\psi} = \hat{\omega}/(4\pi^2|k|^2)$, $H(\hat{\omega}) = |\hat{\omega}|^2/(8\pi^2|k|^2)$.

Theorem 9. *For $k \neq 0$, $k \cdot \hat{u} = 0$: $H(\operatorname{curl} \hat{u}) = \frac{1}{2}|\hat{u}|^2$; $H(\omega + \varepsilon\eta) = H(\omega) + \varepsilon \operatorname{Re}\langle \hat{\psi}, \eta \rangle + \varepsilon^2 H(\eta)$, so $\delta H/\delta\omega = \hat{\psi}$; and $2\pi i\, k \times \hat{\psi} = \hat{u}$, i.e. $\operatorname{curl}(\delta H/\delta\omega) = u$.*

Proof. $k \times (k \times \hat{u}) = k(k \cdot \hat{u}) - |k|^2 \hat{u} = -|k|^2 \hat{u}$, so $(2\pi i)^2 k \times (k \times \hat{u})/(4\pi^2|k|^2) = \hat{u}$. The expansion is the polarisation of a real quadratic form. □

Commentary. The original degeneracy argument read: "2. $\delta H/\delta\omega = u =$ Biot–Savart ω. 3. By definition of Π_u we have $\Pi_u(u) = 0$" (*Tether_May31.lean*, Drive), and the May drafts wrote "$\delta H/\delta\mu = u$ (Biot–Savart recovery, Coulomb gauge)" (*May 16 edits*). Formalising it exposed the slip: as a functional of vorticity, the derivative of the kinetic energy is the stream function ψ, not u, and the ψ-slot tether failed energy degeneracy for that reason. The repair was to put the curl into the tether slot, so that what it needs is $\operatorname{curl}(\delta H/\delta\omega) = u$. In the tether build that identity was a hypothesis, because the derivative there is abstract. In Fourier variables it is the theorem above: the earlier statement becomes true once the curl is put back in.

6 Scope

Only $p = 2$ strain bounds are proved. Nothing here concerns existence or regularity of Navier–Stokes solutions.